

The Locus of Linguistic Knowledge: An Exemplar Model of Phonetic Neutralization

Jonathan Charles Paramore
Washington University in St. Louis
j.paramore@wustl.edu

Abstract

Eliminativist theories of speech production and perception locate linguistic knowledge exclusively within detailed episodic speech representations and their interactions. This paper examines the tenability of that commitment. Through a series of exemplar based computational simulations modeling the contextual neutralization of phonemic vowel nasality over time for an individual speaker, I explore what is required for eliminativist models to predict both phonetically complete and incomplete neutralization. The simulations show that, with the appropriate weighting of competing production biases of anticipatory nasalization and morphological similarity, eliminativist models can successfully produce phonetically incomplete neutralization. While token frequency manipulations and socially weighted encoding of exemplar memories can achieve phonetically complete neutralization, they introduce questionable empirical predictions and over-generate the predicted types of neutralization. Adjusting the relative strength of production biases provides an alternative solution by regulating the degree of neutralization while keeping the production optimum bounded by the competing phonetic targets of production biases. I argue that permitting learned, language-specific production-bias weights provides a more explanatorily adequate account of neutralization than the frequency-based or socially weighted encoding approaches examined here. Crucially, this account requires a form of linguistic knowledge that extends beyond the network of episodic speech representations and their interactions.

1 Introduction

Under an eliminativist exemplar model of speech production and perception (Bybee, 2001), all linguistic information is stored within a complex network of highly detailed and multidimensional speech representations. Rather than attributing phonological productivity to formal rules or constraints that operate over abstract symbolic representations to generate predictable outputs — as is standard in classical generative theories of speech production (Chomsky and Halle, 1968) — eliminativist models

derive productivity from the distribution and associative activation of stored speech memories. Repeated experiences with a particular pattern entrench that pattern in the exemplar representational space, and similarity-based retrieval causes those entrenched patterns to guide both familiar and novel future productions. The result is rule-like regularity across the lexicon: what appears to be the categorical application of a formal rule instead emerges from the cumulative influence of past language experience on present production. A central claim of eliminativist exemplar theories, then, is that a separate form of linguistic knowledge beyond the network of exemplar memories is superfluous. The organization and activation of speech memories is sufficient to explain systematic and productive linguistic patterns.

The typology of phonetic neutralization patterns provides an interesting testbed for the assertion that representations and their relations are the sole capsules of linguistic information. Through a series of computational simulations modeling the evolution of vowel nasality patterns within an individual speaker’s episodic memory network, I show that shifting the neutralization point of a neutralizing category requires a lasting change in the balance of influences governing production. Although manipulating frequency effects (Pierrehumbert, 2001) and/or relying on the socially weighted encoding of exemplar memories (Sumner et al., 2014) can provide the necessary influences to shift the endpoint of phonetic change within an eliminativist framework, I argue that these methods make additional undesirable empirical predictions and theoretical commitments. As an alternative, I propose that treating the relative strength of competing production biases as learned language-specific parameters provides a more explanatorily adequate account of neutralization. Crucially, this proposal locates linguistic knowledge, at least in part, beyond episodic speech representations and their relations.

Over the last several decades, the experimental literature has provided copious examples showing that the phonological neutralization of a contrast can be phonetically incomplete (Bennett and Eischens, 2026; Braver, 2013, 2014, 2019; Braver and Kawahara, 2016; Dinnsen and Charles-Luce, 1984; Du and Durvasula, 2022; Gerfen, 2002; Matsui, 2015; Port and O’Dell, 1985; Warner et al., 2004; Yu, 2007; Zeng et al., 2025). For example, in German, stem-final voiced obstruents are devoiced when they occur at the right edge of a prosodic word, as shown by the examples in (1a).¹ As (1b) shows, the voicing alternation cannot be explained in the opposite direction, in which obstruents are voiced intervocally. Otherwise, we would expect the stem-final stop in ‘council’ to exhibit voicing in the plural form as well.

- (1) Word-final obstruent devoicing in German (Port and O’Dell, 1985)
- a. [ʁa:t] ‘wheel.SG’ vs. [ʁɛ:d-ɐ] ‘wheel-PL’
 - b. [ʁa:t] ‘council.SG’ vs. [ʁɛ:t-ə] ‘council-PL’

¹The vowel umlaut in the plural forms are additional markers of pluralization in German. Also, the distinct plural suffixes across the two lexemes are lexically determined allomorphs. Neither change affects the present discussion.

To explain the neutralization of obstruent voicing word finally in German, eliminativist models draw from the associative architecture of a densely populated and highly redundant representational space, coupled with phonetic production pressures. Potentially voiced obstruents in the word-final position — like [ʁa:t] ‘wheel.SG’ — are biased toward voicelessness due to a phonetically natural production bias to spread the glottis at the end of an utterance in anticipation of a pause. This utterance-final devoicing bias can then generalize to the word-level domain (Myers and Padgett, 2014), resulting in an articulatory preference by speakers to devoice word-final obstruents. Over time, as more and more exemplar memories encoded with word-final voiceless obstruents are stored, word-final obstruent voicelessness emerges as an entrenched rule-like phonological pattern in the lexicon, despite the absence of a formal rule that enforces it.

Moreover, while obstruent voicing is *phonologically* neutralized word-finally, numerous phonetic studies have shown that vowels occurring before voiced obstruents are measurably longer than those occurring before voiceless obstruents in German, even when final devoicing has applied (Braunschweiler, 1997; Port and O’Dell, 1985). Thus, the vowel in [ʁa:t] ‘wheel.SG’ is demonstrably longer than the vowel in [ʁa:t] ‘council.SG’.

Within the eliminativist framework, the measurably longer duration of the vowel in [ʁa:t] ‘wheel.SG’ is attributed to its paradigmatic relationship with [ʁɛ:d-ɐ] ‘wheel-PL’, in which the vowel is transparently longer before the voiced [d] (a common coarticulatory effect distinguishing voiced and voiceless obstruents, Coretta 2020). When producing [ʁa:t] ‘wheel.SG’, exemplars in the singular form, as well as paradigmatically related plural exemplars, are activated. These associative activations influence production, leading to slightly longer vowels in the singular form. However, for [ʁa:t] ‘council.SG’, no corresponding attraction toward a longer vowel exists because the plural form contains a voiceless intervocalic obstruent: [ʁɛ:tə].

Phonetic neutralization of this kind is not limited to word-final obstruent voicing patterns. Similar incomplete neutralization patterns have been reported for vowel nasality contrasts in San Martín Peras Mixtec (SMPM) (Bennett and Eischens, 2026), for instance. In SMPM, vowel nasality is contrastive following an oral consonant (CV vs. C $\tilde{V}$), but this contrast is neutralized following a nasal consonant (N $\tilde{V}$ but *NV). Crucially, the realization of nasality on post-N vowels is weaker than the nasality on contrastive C $\tilde{V}$ vowels, such that a measurable and perceptually salient phonetic distinction exists between predictably nasal N $\tilde{V}$ vowels on the one hand, and both contrastive oral CV vowels and contrastive nasal C $\tilde{V}$ vowels on the other hand.

Despite rampant evidence of phonetically incomplete neutralization, other studies have shown that the phonological neutralization of a contrast can also be phonetically *complete* (Ellis and Hardcastle, 2002; Kim and Jongman, 1996; Kochetov and Pouplier, 2008; Lahiri et al., 1987; Paramore, 2026; Vietti et al., 2018). In the endangered Pakistani language of Mankiyali, for instance, Paramore (2026, ch.4) describes a vowel system, in which oral and nasal vowels are contrastive, as shown in (2).

- (2) Contrastive vowel nasality in Mankiyali
- a. [d̥ɪ:] ‘giant.NOM’ vs. [d̥ĩ:] ‘pine firewood.NOM’
 - b. [gɑ:] ‘cow.NOM’ vs. [gã:] ‘village.NOM’

When followed by a nasal consonant suffix like the genitive case marker in (3), however, this contrast is neutralized, such that the otherwise contrastive vowels are both produced as nasal vowels. Importantly, Paramore (2026, ch.4) demonstrates that this neutralization process is not only phonologically complete, but phonetically complete as well: no measurable phonetic trace exists to distinguish the two typically contrastive vowels from one another.

- (3) Neutralization of vowel nasality contrasts in Mankiyali
- a. /d̥ɪ:ŋ/ → [d̥ĩ:ŋ] ‘giant-GEN.F.SG’
 - b. /d̥ĩ:ŋ/ → [d̥ĩ:ŋ] ‘pine firewood-GEN.F.SG’
 - c. /gɑ:ŋ/ → [gã:ŋ] ‘cow-GEN.F.SG’
 - d. /gã:ŋ/ → [gã:ŋ] ‘village-GEN’

The attestation of both phonetically complete and phonetically incomplete neutralization raises an important question for eliminativist models: How can an eliminativist system explain complete neutralization while retaining the paradigmatic influences needed to explain incomplete neutralization? By simulating an individual language user’s production and storage of exemplar memories over time through a series of models constructed from an eliminativist exemplar framework, I show that, while changes to the phonetic substance of exemplar memories within a neutralizing category may alter the trajectory of sound change within that category, it does so without durably shifting its final stable state under otherwise fixed conditions. Ultimately, I suggest that effectuating a change in the stable state of a neutralized category within the eliminativist architecture has unintended consequences and propose an alternative solution which permits the locus of linguistic knowledge to extend beyond the network of speech representations.

2 Overview of Exemplar Dynamics

Eight core principles relevant for the current discussion form the basis of eliminativist exemplar frameworks of linguistic cognition: redundant representations, similarity-based association, frequency effects, emergent structure, production biases, decay, differential encoding, and social indexing. Each of these principles is discussed in more detail below. Eliminativist theories are unified by the shared assumption that language use — as opposed to an innate formal grammar — is the main driver of linguistic cognition. The eliminativist model outlined here is based primarily on exemplar-theoretic models proposed in Bybee (2001), Pierrehumbert (2001), Wedel (2004), and Kaplan (2015), but these ideas are held by most proponents of eliminativist frameworks.

2.1 Redundant Representations

Whereas classic generative theories treat mental representations of morphemes as abstract symbolic categories with a single representation that omits predictable phonetic content, eliminativist exemplar theories maintain that speakers store memories of individual sounds, whole words, and sometimes phrases (Wedel, 2004, p.33). Moreover, the phonetic detail associated with an individual exemplar memory is a central component of these representations, rather than being discarded as an irrelevant redundancy of phonetic implementation. Rather than stripping token-specific details present in an individual experience from a representation, exemplar theories treat the fine-grained phonetic details and the social characteristics of the speaker who uttered the token as the core substance of the representation.

Moreover, representations of words and phrases do not consist of a single form representing an entire category, but rather an interconnected ‘cloud’ of richly detailed exemplars that collectively contribute to a speaker’s stored knowledge of a linguistic form.

2.2 Similarity-Based Association

Individual exemplar memories that are semantically and/or phonetically similar are strongly linked in memory (Bybee, 2001, p.22-23). The stronger these ties, the greater the influence an exemplar imposes on the future production patterns of its related forms.

However, Kaplan (2015) demonstrates that the influence exerted by paradigmatically related forms cannot be symmetric. In a series of exemplar-based simulations modeling positional neutralization through word-final devoicing, Kaplan implemented both (i) a production bias favoring devoicing in word-final position and (ii) a paradigmatic similarity bias that caused morphologically related forms to influence one another equally. Under this symmetric similarity bias, non-neutralizing forms containing voiced obstruents in non-final position (e.g., [gibi]) were gradually pulled toward the devoiced variants found in paradigmatically related neutralizing forms (e.g., [gip]), eventually losing the voicing contrast altogether. As a result, obstruent devoicing spread throughout the entire paradigm rather than remaining restricted to word-final position.

However, when Kaplan modified the simulations so that paradigmatic influence was asymmetric — such that neutralizing forms were influenced by non-neutralizing forms, but not vice versa — the simulations successfully produced stable positional neutralization patterns in which voicing contrasts were preserved in non-final contexts while neutralization remained restricted to word-final position.

If the paradigmatic influence between neutralized and non-neutralized forms is asymmetric, as argued by Kaplan (2015), we should only expect to see the influence of non-neutralized forms on related neutralized forms. Under this view, lexical associations between neutralizing and non-neutralizing forms give rise to incomplete neu-

tralization, as observed for non-neutralized voiced stop in German words like [ʁɛːdɐ] ‘wheels’ exerting influence on the neutralized final devoiced stop in [ʁaːt] ‘wheel’.

2.3 Frequency Effects

Because exemplars accumulate over time, speech patterns within a given lexical item are primarily shaped by the highest concentration of exemplars — the center of gravity — within that lexical item. As more exemplars accumulate around this center of gravity, their specific phonetic details collude to direct the production patterns of future exemplars, resulting in lexical *entrenchment* (Pierrehumbert, 2001, p.149-152), in which a specific phonological pattern is reinforced and eventually cemented within an exemplar cloud, barring external pressures from related processes or exemplar clouds.

Entrenchment also takes effect at a more general ‘phonological’ level across related forms in the lexicon. Because word forms that are used more frequently accumulate exemplars more rapidly than less frequent forms (i.e., a higher token frequency), the sound patterns present in frequent forms direct the evolution of the exemplar network as a whole, to the effect that the patterns present in high-frequency forms permeate all relevant forms in the lexicon over time (Wedel, 2004).

2.4 Emergent Structure

Exemplar frameworks contend that the richly detailed and interconnected nature of the lexicon is sufficient to explain the presence of regular phonological patterns in a lexicon. That is, instead of positing the existence of a formal grammar through which representations are filtered to generate systematic phonological patterns across the lexicon, phonetic patterns that are repeated with high degrees of frequency across exemplars and forms generate the appearance of a formal phonological rule, when in reality the pattern is an emergent property of the exemplar network itself (Bybee, 2001, p.3-4). In this view, then, grammar and representations are inseparable; there is no transcendent locus of abstract linguistic knowledge beyond individual exemplar memories and their relations.

2.5 Production Biases

According to Pierrehumbert (2001, p.147), historical sound change occurs when production biases are introduced into a language. These biases can take the form of lenition of a certain sound, or, particularly relevant for the simulations in this paper, nasal coarticulation on a preceding vowel. When such changes are permitted, they induce gradual shifts in the distribution of exemplar clouds over time. Ultimately, regardless of how long a category takes to arrive at a new stabilized production point,

historical sound changes converge on a specific equilibrium state at which production biases are optimally satisfied.

2.6 Decay

Memories are not permanent; if not strengthened through repetition, they fade and decay over time (Pierrehumbert, 2001, p.140-141). Exemplar memories residing in the outskirts of an exemplar cloud distribution are especially susceptible to decay due the lack of nearby exemplars to reinforce their activation. Decay is clearly affected by recency as well, in which exemplars heard or produced in the more distant past are more likely to decay.

2.7 Encoding Strength

Not all individual exemplar memories are encoded with equal strength. Various factors such as recency, location in the representational space, and speaker identity result in differential encoding strengths, in which some exemplars exhibit a higher level of ‘resting activation’ than temporally distant exemplars occupying sparser areas of the representational space (Pierrehumbert, 2001, p.141). As a result, these strongly encoded exemplars are more easily retrieved in the speech production process and exert greater influence on future productions.

2.8 Social Information

Speakers adapt their speech production patterns for particular social settings, utilizing different phonetic patterns in different contexts. As such, they must carve up the exemplar space based on social information. This suggests that exemplar memories are not only encoded with detailed phonetic content, but they are also indexed with relevant social information.

In her model of exemplar dynamics, Pierrehumbert (2001, p.145-152) suggests that ‘attentional weighting’ must be an additional factor to explain the effects of social context on speech production patterns. As we will see, if higher-order linguistic knowledge is rejected, the *socially weighted encoding* (Sumner et al., 2014) of individual exemplars becomes crucial for explaining how different types of neutralization arise, though such an approach is not without its consequences.

3 Constructing a model of sound change

The main function of the following simulations is to demonstrate how the foundational tenets of eliminativist frameworks might account for the contextual neutralization of

a vowel nasality contrast.² Constructed from the principles of eliminativist theories of speech production and perception outlined above, the models demonstrate that a speaker’s nasality patterns converge toward a stable state of nasality that maximally satisfies the competing production biases exerting pressure on the exemplar space. Although different starting points of nasality may alter the *rate* and *path* toward convergence, the model consistently settles on the same nasality state that optimally satisfies the pressures from production biases. Only changes linked to something other than the phonetic content of individual exemplars can counteract the stabilizing pressure induced by production biases.

The models take as an initial input, words from four inflectional categories with pre-specified frequency distributions. Exemplars accumulate within each category over time based on these frequencies through an iterative process of exemplar creation, optimization, and storage. The output of each model is a plot showing the evolution of nasality within these four categories over time and the stable state that each category arrives at. The Python code for each model presented here can be found at [author’s_github](#).

3.1 Model Input

The four input categories are comprised of two ‘non-neutralizing’ forms with contrastive oral (ORAL) and contrastive nasal (NASAL) vowels respectively, and two ‘neutralizing’ forms inflected with a nasal suffix (ORAL-N and NASAL-N). As summarized in Table 1, each of these four categories contains two lexical items, yielding eight distinct word-level exemplar clouds. To simplify the exposition, I focus on a single phonetic property of these word categories: nasality. In reality, exemplar categories for words exist in a high-dimensional space, with a large number of phonetic factors influencing the shape of the cloud. I set these complexities aside in what follows.

Oral	Nasal	Oral-N	Nasal-N
ta:	tã:	ta:-n	tã:-n
ki:	kĩ:	ki:-n	kĩ:-n

Table 1: The model input consists of four categories with two lexical items each, for a total of eight exemplar clouds.

At initialization, each of the eight clouds is seeded with a single exemplar, the substance of which is a single scalar value bounded between 0 and 1 that corresponds to the mean nasality of the exemplar’s vowel. Nasality values near 0 correspond to minimal or absent nasality, whereas values near 1 indicate a completely nasal segment. Unless otherwise noted, the initial exemplars in the two oral categories are randomly

²Morley (2023) develops a similar class of models to the ones outlined here, and her results implicitly entail the need for mechanisms external to the phonetic content of exemplar representations in order to achieve different stable states in an exemplar system.

drawn from a normal distribution with $\mu = .2$ and $\sigma = .025$, while the initial exemplars in the two nasal categories are randomly drawn from a normal distribution with $\mu = .6$ and $\sigma = .025$. These distributions approximate the baseline nasalance values of oral and nasal vowels from previous phonetic studies measuring nasalance (e.g., Bennett and Eischens 2026; Paramore 2026, ch.4). Thus, the initial state of nasality in the models lack any neutralization of nasality in the neutralizing ORAL-N and NASAL-N forms.

Finally, the relative frequency of the lexical items is also specified at model initialization. These frequencies determine the probability that a given lexical item is selected to generate a new exemplar during each iteration of the production process. Unless otherwise noted, selection probabilities are uniform across the eight lexical items, with each item having a .125 probability of being chosen to generate a new exemplar in a given iteration.

3.2 Exemplar creation, optimization, and storage

To accurately model the core principles of eliminativist theories outlined in section 2, the exemplar accumulation process proceeds in six ordered steps: (1) selection, (2) entrenchment, (3) optimization, (4) noise, (5) storage, (6) and decay. These steps are repeated 1,000 times in a single simulation to model potential changes in the distribution of exemplar clouds over time through the accumulation of 1,000 exemplar productions. To clearly demonstrate the effect of the production biases on the stable end state of each category in the model, I begin with a minimalist model and slowly build up to a full-fledged model that incorporates all the necessary eliminativist principles. Section 4 demonstrates how the basic principles of eliminativist models achieve incomplete neutralization, and section 5 considers how such models might also achieve complete neutralization.

4 Modeling Incomplete neutralization

4.1 Model 1: selection, entrenchment, noise, storage, decay

The initial model includes the selection of a lexical item for the creation of a new exemplar, the entrenchment of that new exemplar’s nasality value within its lexical category, the addition of noise, storage of that exemplar in the appropriate cloud, and the decay of older exemplar memories.

For each iteration, selection occurred by randomly choosing one of the four categories (ORAL, ORAL-N, NASAL, or NASAL-N) and then randomly selecting one of the two lexical forms within that category for classifying the newly generated exemplar.

After the specific lexical form is selected for classification of the new exemplar, an entrenchment mechanism is employed to capture the tendency for speakers to progressively concentrate their neuromotor productions on the most practiced, frequently

repeated patterns within a category’s distribution (Bybee, 2001, p.54). To model this, new exemplars are initially assigned the mean nasality of their lexical form’s cloud. The mean provides a good approximation of a cloud’s most typical nasality value, and as additional exemplars accumulate, this mean becomes increasingly stable in the face of extreme productions. In effect, the center of gravity of each lexical form becomes progressively ‘cemented’ as the simulation proceeds, and newly generated exemplars reinforce this central value. Only a systematic production bias, overwhelming frequency changes in related forms, or changes in the activation strength of related exemplars are capable of shifting a category away from its entrenched mean.

After assigning the new exemplar with its cloud’s mean nasality value, random noise ($\sigma = .025$) was applied to the nasality value to account for inherent imprecision and variability in the production of a nasality target. The new exemplar is then stored in the appropriate lexical cloud.

To account for the fact that memories are not permanent, the model imposes a memory capacity limit of fifty exemplars per lexical form. Once a lexical form accumulates 50 exemplars, the generation of newly created exemplars in that lexical form’s cloud triggers the removal of the oldest exemplar in the cloud, ensuring that it no longer contributes to the lexical form’s mean estimations, either during baseline nasality assignment within an iteration or in the final mean calculation of the lexical item at the end of a simulation. Under normal model conditions, in which all lexical forms have an equal probability of being selected for production, each lexical cloud accumulates approximately 125 exemplars over the course of a single simulation. This means that the ~ 75 earliest exemplars for each form decay by the end of the simulation. This decay mechanism also serves to further reinforce entrenchment of high-frequency patterns in the lexicon: early low-nasality exemplars in a shifting category like ORAL-N lose their influence in shaping the mean as more entrenched exemplars take their place in the cloud.

Figure 1 shows the output of a model with the above specifications after 1,000 iterations. While the model shows the accumulation of 1,000 exemplar productions across eight lexical clouds, the nasality of all exemplars within each cloud are relatively stable. This stability arises because each cloud is initialized with a single exemplar, and the nasality of all subsequent exemplars in that cloud have no pressure to deviate from this value. As a result, each lexical form’s exemplars exhibit a relatively static nasality across the 1,000 iterations.

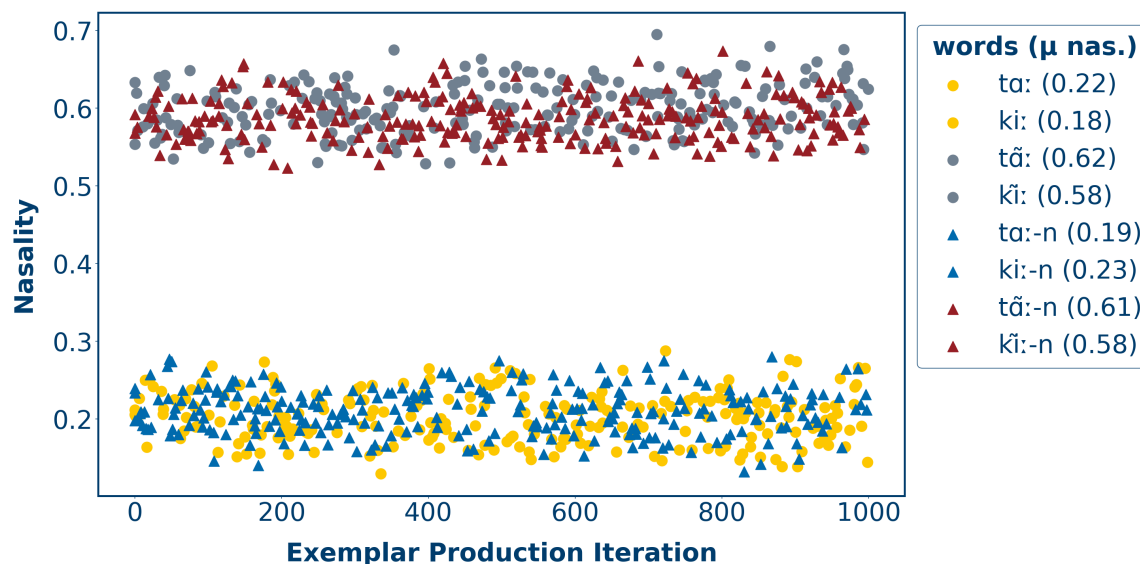


Figure 1: Model 1: selection, entrenchment, noise, storage, decay. The plot shows nasality values of individual exemplars over 1,000 iterations. Non-neutralizing ORAL exemplars are in yellow, non-neutralizing NASAL exemplars in grey, neutralizing ORAL-N exemplars in blue, and neutralizing NASAL-N exemplars in red. Values in the key show mean nasality of each lexical item’s cloud at the end of the simulation.

4.2 Model 2: selection, entrenchment, pre-N bias, noise, storage, decay

To achieve a pressure toward neutralization, a production bias pressuring pre-N vowels to be more nasalized is needed.³ As such, after a newly created exemplar’s lexical identity is selected and its initial nasality entrenched as the mean of that cloud, a pre-N bias favors an earlier onset of velar lowering during the pre-N vowel in anticipation of the following nasal consonant. This bias reflects a production ease pressure: initiating velum lowering during the vowel reduces the need for a rapid articulatory shift at the onset of the nasal consonant. In the model, this is encoded as a penalty for insufficient nasality in pre-N vowels.

I implement this bias through the minimization of an objective function by gradient descent using `scipy.optimize` in Python. In the current model, the objective function value is computed solely by determining the deviation of an exemplar’s nasality from a typical nasal vowel in the NASAL category. As shown by the equation in (4), the penalty is calculated by squaring the difference between the nasality of the newly

³Morley (2023) develops another approach to implementing production biases. In her models, the bias and the final stable state (what she calls the target) are distinct mechanisms. Here, however, my implementation combines the production bias and the target into a single component. Specifically, the bias is implemented as a penalty for deviating from a certain phonetic value, and the target is the point at which the bias penalty is minimized.

created exemplar and the mean nasality of exemplars in the NASAL category. In effect, the penalization of the newly created exemplar will be quadratically larger as its nasality value diverges from that of a typical vowel in the NASAL category.⁴ Crucially, because the pre-N production bias only applies to forms with a pre-N vowel, the objective function value will be zero for exemplars selected from either the non-neutralizing ORAL or NASAL categories. That is, there is no pressure to lower the velum in anticipation of an upcoming nasal consonant in these forms, so the bias does not apply. For both ORAL-N and NASAL-N exemplars, on the other hand, the bias is applicable, albeit somewhat vacuously for NASAL-N exemplars.

$$(4) \quad \text{Pre-N Nasality Bias} \\ (NASAL\mu_{nas} - exemplar_{nas})^2$$

An additional restriction on the optimization process involves the limitation of changes in an exemplar’s nasality to $\pm .075$ for a single iteration. Without such a restriction, the optimization function would instigate a dramatic shift in nasality at the outset of the simulation to maximally satisfy the pre-N nasality bias in (4), resulting in instant changes in both pre-N category’s nasality values toward the mean nasality of the NASAL category. In reality, sound change is gradual, and effects on individual exemplars are only slight. Capping changes in nasality for an individual exemplar to $.075$ prevents one-fell-swoop changes and more closely imitates the gradual shift of a speaker’s productions over time.

Figure 2 shows the results of incorporating the pre-N bias into the model. Clearly, the ORAL-N exemplars — pressured by the pre-N bias — exhibit an upward trajectory toward the nasality space occupied by the NASAL and NASAL-N categories. Once the ORAL-N mean reaches nasality levels typical of a nasal vowel, the category stabilizes, and the stabilized nasality pattern becomes increasingly entrenched around the optimized mean.

⁴There are several advantages to defining production biases as squared deviations. First, squared deviations are continuously differentiable, which makes them well-suited for optimization via gradient descent. By contrast, an absolute-deviation formulation — $|NASAL\mu_{nas} - exemplar_{nas}|$ — yields a piecewise linear objective function whose derivative is undefined at the optimum and constant elsewhere, which often leads to less stable convergence behavior. Second, squared deviations introduce curvature into the objective function, which guarantees a unique interior minimum when multiple biases are in conflict. This property is crucial for modeling incomplete neutralization (section 4.3), as it allows the optimal value (i.e., the stabilization point in the system) to fall between competing targets rather than necessarily collapsing to the bias with the highest weight. In contrast, absolute-deviation penalties tend to produce boundary solutions (i.e., selection of one target over another) unless the competing weights are exactly equal, in which case the objective is flat between the targets. Finally, squared deviations penalize larger departures from the target more strongly than smaller ones, reflecting the theoretically grounded intuition that exemplars that are substantially misaligned with a bias should incur disproportionately greater cost than those that deviate only minimally.

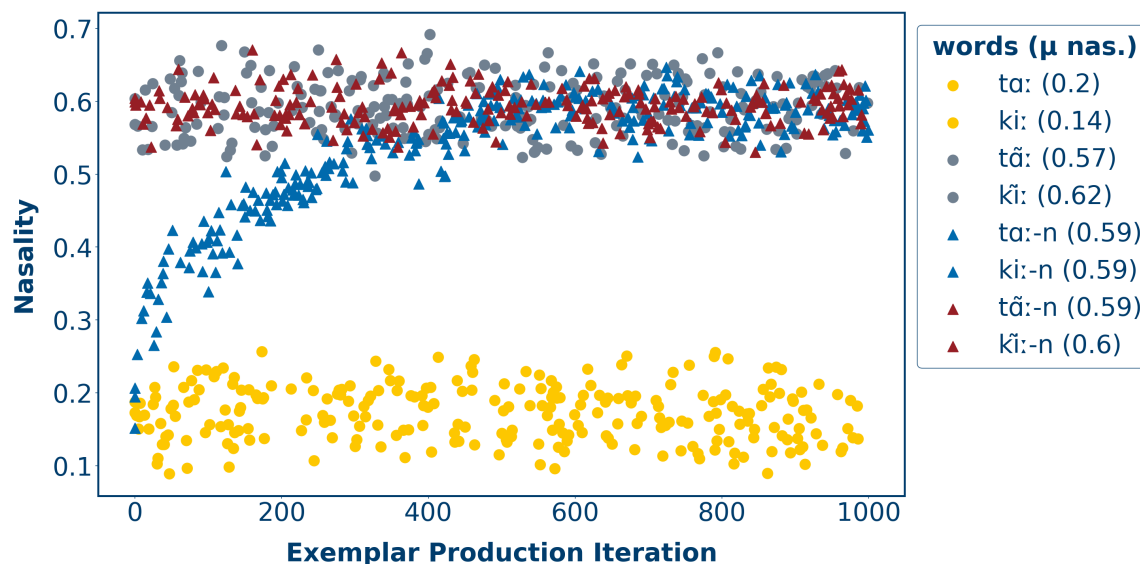


Figure 2: Model 2: selection, entrenchment, pre-N bias, noise, storage, decay.

However, as the similarity-based association effects described in section 2.2 have not been included in the model, this is not a surprising result. No bias currently exists to counteract the steadily increasing nasality of the ORAL-N category. In the next model, we will see that the introduction of a similarity bias — in which non-neutralizing forms attract their paradigmatically related neutralizing forms — works to prevent complete neutralization of vowel nasality in the ORAL-N category.

4.3 Model 3: selection, entrenchment, pre-N/similarity biases, noise, storage, decay

A fundamental predictor of sound change, neutralization patterns, and many other related processes under eliminativist frameworks is the interconnectedness of phonetically and semantically similar exemplars *across* lexical forms (see section 2.2). In these approaches, the production of a new exemplar is shaped not only by the exemplar memories belonging to its own lexical cloud, but also by paradigmatically-related exemplar clouds. Crucially, this paradigmatic similarity effect is asymmetric, in that exemplars of non-neutralizing forms in a paradigm unidirectionally attract exemplars of neutralizing forms (Kaplan, 2015).

To incorporate these similarity effects into the simulations of *Model 3*, a morphological similarity bias was added to the objective function. As formalized in (5), the similarity bias penalizes exemplars of neutralizing forms (i.e., ORAL-N and NASAL-N) that deviate from the mean nasality of their related non-neutralizing form, α (i.e., ORAL and NASAL). Like the pre-N nasality bias in (4), the similarity bias penalty increases quadratically as discrepancy between a new exemplar and the non-neutralizing

forms (α) in its paradigm grows (See footnote 4 for the advantages of constructing biases in this way).

$$(5) \quad \text{Morphological Similarity Bias} \\ (\alpha\mu_{nas} - exemplar_{nas})^2$$

The objective function that integrates both biases is provided in (6). In *Model 3*, the pre-N bias is modulated by a scalar of five. This scalar amplifies the influence of the pre-N bias relative to the morphological similarity bias. Without it, the two pressures would be roughly equal in magnitude but point in opposite directions for ORAL-N exemplars, producing an equilibrium midway between the ORAL and NASAL categories. By introducing this scalar, the pre-N bias exercises an outsized influence in the optimization process, yielding higher degrees of nasality neutralization as the scalar increases.

$$(6) \quad \text{Objective Function of Model 3}$$

$$\mathcal{O}_{Model3} = \underbrace{(\alpha\mu_{nas} - exemplar_{nas})^2}_{\text{Morphological Similarity Bias}} + 5 \times \underbrace{(NASAL\mu_{nas} - exemplar_{nas})^2}_{\text{pre-N Nasality Bias}}$$

During the optimization stage of the simulation, the nasality value of a newly created exemplar is adjusted to minimize the objective function in (6). At early stages of the accumulation process, ORAL-N exemplars differ only slightly from the ORAL forms in their paradigm, making the similarity bias term very small. In contrast, the difference between early ORAL-N exemplars and underlying nasal vowels is large, so the pre-N term dominates the objective function. Consequently, early ORAL-N exemplars undergo substantial upward shifts in nasality while the model seeks to reduce the overall penalty.

Over many iterations, these increases gradually shift the mean of the ORAL-N distribution upward toward the nasality level of the NASAL and NASAL-N categories. As this occurs, the pre-N bias term decreases, while the similarity bias term penalty, reflecting the growing distance from the non-neutralizing ORAL form, increases. The interaction between these two terms behaves like children on a seesaw: as one term falls the other rises. However, because the pre-N bias is multiplied by a scalar, the equilibrium is not reached where the two squared distances are equal, but instead at a point where the weighted pull toward the nasal category balances the pull back toward the non-neutralizing form. Just as a heavier child on a seesaw rests closer to the ground, the stable point for ORAL-N tokens settles closer to the mean of the underlying nasal categories.

The optimal nasality value of ORAL-N forms under the weighted objective function in (6) is illustrated in Figure 3. The green solid and pink dotted curves plot the individual biases that make up the objective function, and the thick navy curve

shows their summed penalties. The objective function is maximally satisfied at the nasality value that minimizes the entire function, as delineated by the blue vertical dashed line labeled ‘optimal ORAL-N’.

Increasing or decreasing the scalar of the pre-N bias shifts the optimal nasality value of the objective function accordingly. Notably, because the similarity bias attracts exemplars toward the mean of the ORAL category (yellow dashed line) and the pre-N bias attracts them toward the NASAL mean (gray dashed line), the stabilization point is necessarily bounded between these two values.

This provides an important prediction: ‘over-neutralization’ — in which a neutralizing segment like an ORAL-N vowel *surpasses* the phonetic value of underlying category that attracts it — is not expected to occur under the current model parameters. As Du and Durvasula (2025, p.21) observe, over-neutralization appears unattested, and the present model accurately avoids predicting over-neutralization as a possible stable state.

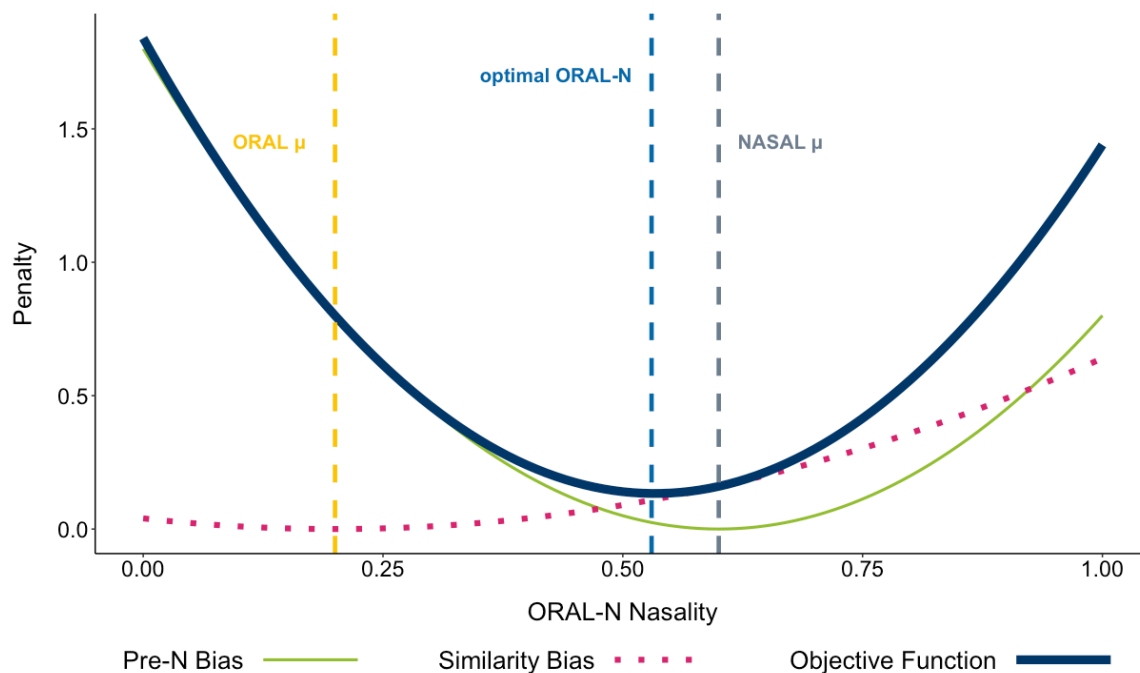


Figure 3: *Model 3* objective function: penalty landscape for ORAL-N forms as determined by the two bias components that make up the objective function. The solid green curve shows the pre-N bias when scaled by 5; the dotted pink curve shows the similarity bias. The thick navy curve plots their combined objective function, with its minimum corresponding to the optimal nasality target for ORAL-N exemplars. Vertical dashed lines mark the typical ORAL mean (yellow) at 0.2, the typical NASAL mean (gray) at 0.6, and the resulting optimal ORAL-N value (blue). The figure illustrates how competing pressures jointly determine an intermediate nasality target for ORAL-N forms.

A representative simulation of *Model 3* is shown in Figure 4. As is evident, exemplars in the ORAL-N category begin with nasality similar to ORAL vowels but over time increase in nasality until reaching a plateau and stabilizing just below the NASAL and NASAL-N categories, thus achieving phonetically incomplete neutralization.

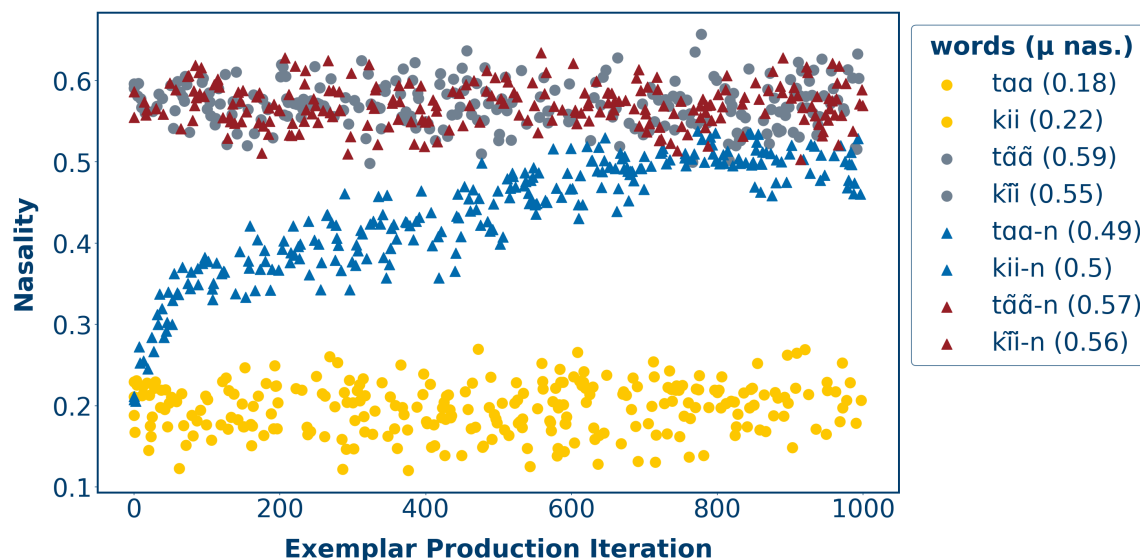


Figure 4: Model 3: selection, entrenchment, pre-N/similarity biases, noise, storage, decay.

To demonstrate that stabilization of the ORAL-N category in a state of phonetically incomplete neutralization, as presented in Figure 4, is consistent across trials, I conducted 100 independent simulations using the parameters of *Model 3* and recorded the final mean nasality of each lexical category at the end of each simulation. The dot and whisker plot in Figure 5 summarizes the distribution of these endpoint means by plotting the mean nasality and 95% confidence intervals for each of the four word categories' final nasality means. As the plot shows, across simulations the ORAL-N category consistently approaches, but remains phonetically distinct from, the underlying nasal categories, with non-overlapping confidence intervals. Under the parameters of the current model, then, an eliminativist model accurately predicts phonetically incomplete neutralization.

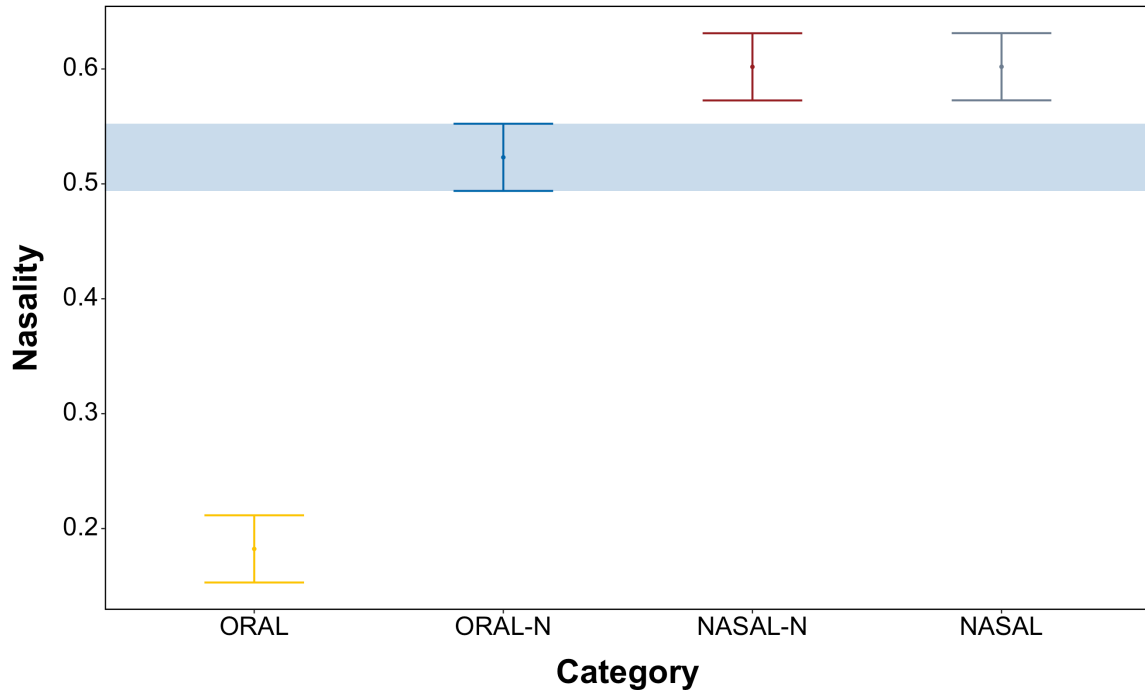


Figure 5: Model 3: selection, entrenchment, pre-N/similarity biases, noise, storage, decay. The dot shows the average of the final mean nasality of the four word categories in the model over 100 simulations. Tails represent 95% confidence intervals around the means.

5 Modeling Complete Neutralization

With the core eliminativist architecture mechanisms implemented in *Model 3*, it is clear that eliminativist frameworks naturally predict phonetically incomplete neutralization. The following models explore how this framework might also account for phonetically *complete* neutralization, as observed in the neutralization of the vowel nasality contrast in Mankiyali (Paramore, 2026, ch.4).

Three potential strategies are available to simulate complete neutralization. One approach capable of effectuating complete neutralization that falls squarely within the eliminativist approach involves attenuating the similarity bias strength indirectly through token frequency effects. In section 5.1, I argue that while complete neutralization can be achieved by manipulating frequency effects, the frequency rates of different inflectional categories required to do so are empirically unrealistic.

An alternative approach to attaining complete neutralization within an eliminativist framework again overcomes the pull of the cumulative effects of production biases, but in this case it does so by implementing the socially weighted encoding of individual exemplars (Sumner et al., 2014). This results in exemplar memories ex-

erting non-equal pressure on the outcome of future productions, based on the social characteristics of the talker indexed to the exemplar (section 5.2). However, while the simulations demonstrate that socially weighted encoding effects can produce phonetically complete neutralization, the approach requires a view of memory in which certain representations are completely impervious to decay. Moreover, instituting socially weighted encoding as the sole driver of change also sacrifices the well-established generalization that sound change disproportionately follows articulatorily and perceptually natural pathways, which leads to over-generalizations of the possible types of phonetic neutralization patterns.

A third approach is to relax the eliminativist restriction that linguistic knowledge resides entirely within episodic representations and their relations. Under this approach, a grammatical mechanism external to the exemplar representations regulates the relative weights of the competing production biases directly, allowing the system to stabilize at different phonetic endpoints without fighting against natural production biases. In the model framework outlined in this paper, increasing the scalar on the pre-N nasality bias makes low nasality more costly for ORAL-N vowels, thereby shifting the optimum of the objective function further toward the NASAL category mean. The bias weights thus constitute an additional locus of linguistic knowledge. I argue that permitting this additional form of linguistic knowledge offers a more satisfactory account of the neutralization typology than either the frequency-based or socially weighted encoding approaches.

5.1 Model 4: Complete Neutralization via Frequency Effects

Manipulating token frequency counts can achieve phonetically complete neutralization within the current model. Specifically, reducing the frequency of non-neutralizing ORAL forms reduces the strength of the similarity bias that penalizes high-nasality ORAL-N exemplars. When fewer non-neutralizing ORAL exemplars exist in the network, their pull on the ORAL-N category weakens proportionally. This follows from the basic assumption that high-frequency categories exert disproportionate influence in an exemplar network, while low-frequency categories exert comparatively little influence due to the sparse episodic evidence they contribute. In this way, frequency functions as an implicit scalar of production biases in an eliminativist framework. As the token frequency of a category decreases, the attraction strength of the associated bias term decreases with it.

To implement an indirect effect of frequency-based bias manipulation in the simulation, I initialize *Model 4* identically to *Model 3*, but with the probability of selecting non-neutralizing ORAL forms for exemplar generation reduced to one-tenth the probability of selecting lexical items from the other three categories. As a result, as the simulation proceeds, the influence of the ORAL category diminishes relative to the growing exemplar populations of the other categories. The accumulation trajectory in Figure 6 demonstrates that, under these frequency conditions, an eliminativist

exemplar model can successfully produce phonetically complete neutralization, with ORAL-N forms converging on the nasality of the underlying NASAL categories.

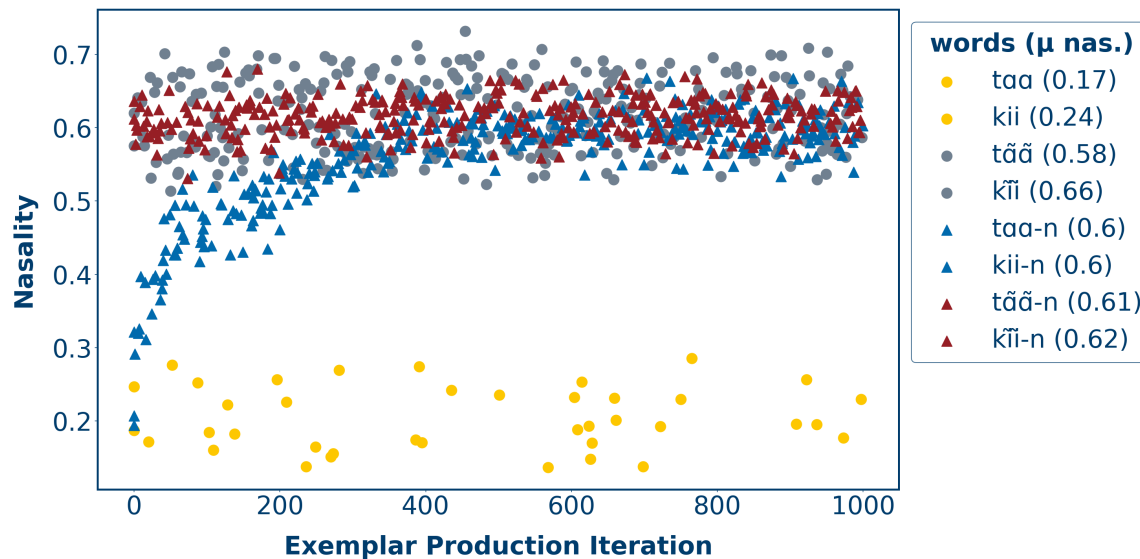


Figure 6: Model 4: selection (low ORAL freq.), entrenchment, pre-N/similarity biases, noise, storage, decay.

Figure 7 confirms that reducing the token frequency of non-neutralizing ORAL forms substantially and reliably weakens the similarity bias opposing neutralization. Across 100 simulations, the ORAL-N category converges with the NASAL and NASAL-N categories, producing a level of nasality consistent with phonetically complete neutralization.⁵

⁵Given the objective function in (6), true complete neutralization is mathematically impossible. Because the function is a weighted sum of squared distances to two distinct attraction points (the non-neutralizing ORAL mean and the NASAL mean), its minimum must always fall strictly between these points. A quadratic attractor function of this form can only reach one endpoint if the weight on the opposing attraction point is exactly zero. However, as the ORAL-N category approaches the NASAL mean, the two distributions may eventually become statistically indistinguishable, which may underlie empirical reports of complete neutralization. Alternatively, once two categories become sufficiently perceptually similar, they may collapse into a single category altogether. This process could be modeled by incorporating gradient attraction dynamics into the model such as those proposed in Burzio (2000).

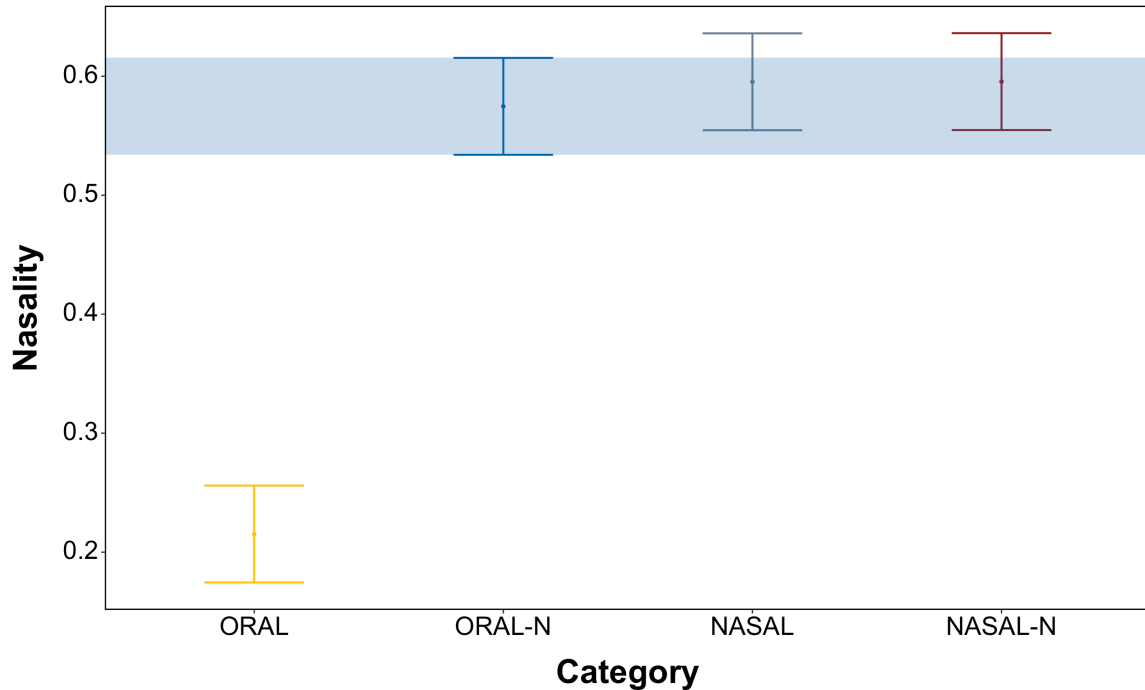


Figure 7: Model 4: selection (low ORAL freq.), entrenchment, pre-N/similarity biases, noise, storage, decay.

While manipulating the token frequency of non-neutralizing ORAL forms successfully produces complete neutralization in the model, this mechanism is unlikely to be a natural or general explanation for complete neutralization in real languages. There are two reasons for this. First, this approach predicts that in any language exhibiting complete neutralization of a feature (e.g., vowel nasality in Mankiyali), non-neutralizing forms across all paradigms that participate in the neutralizing alternation must be substantially less frequent than their corresponding neutralizing forms. Whether such a distributional asymmetry is attested remains an empirical question, but it seems improbable that it would consistently hold across languages and morphological paradigms that exhibit complete neutralization.

To take Mankiyali as a concrete example, no formal corpus exists from which to extract exact token frequencies, but it is unlikely that all non-neutralizing ORAL nouns such as /qɪ:/ ‘giant.NOM’ are markedly rare. On the contrary, the nominative form occurs in a wide range of syntactic environments: as the agent of transitive verbs and the subject of intransitive verbs in the imperfect aspect, and as the patient of transitive verbs and the subject of intransitive verbs in the perfect aspect (Englert, 2022, p.8). By contrast, the primary inflection involving a nasal suffix is the genitive case marker in /qɪ:ŋ/ ‘giant-GEN.F.SG’. It is therefore unlikely that the non-neutralizing forms participating in the neutralizing alternation are consistently low-frequency in the way predicted by the present model.

Relatedly, a frequency-based account of complete neutralization also implies a direct correlation between non-neutralizing-form frequency (e.g., ORAL) and the type of neutralization observed. That is, if complete neutralization results from low-frequency non-neutralizing forms, then languages with phonetically complete neutralization of a particular feature should systematically exhibit unusually low-frequency non-neutralizing forms in relevant paradigms. Conversely, languages exhibiting phonetically incomplete neutralization should exhibit comparatively higher non-neutralizing-form frequencies. These are indeed clear and testable predictions, but it is far from obvious that real languages validate them.

Given the questionable empirical predictions of the frequency-based account, consideration of alternative explanations for complete neutralization is well-motivated.

5.2 Model 5 - Complete Neutralization via Socially Weighted Encoding

Another approach to achieving complete neutralization within the eliminativist framework is to modulate the encoding strength of individual exemplars to the effect that high-nasality exemplars carry more weight in shaping future productions. A natural approach to implement this differential encoding strength is through socially weighted encoding (Eckert, 2000; Eckert and Labov, 2017; Sumner et al., 2014), in which exemplars indexed to talkers viewed by the language user as more socially prestigious are more strongly encoded in memory. By permitting encoding strength differentials, if a set of high-nasality exemplars indexed to a ‘linguistic icon’ (Eckert, 2000) were injected into the system, this small set of influential exemplars could unilaterally determine the nasality of new exemplars, thereby overcoming the fixed stable state wrought by the production biases.

The first step in implementing a socially weighted encoding approach is to inject high-nasality exemplars into the system. These high-nasality exemplars represent an individual’s mimicking of ORAL-N forms with unusually high nasality values compared to his normal productions. This scenario is exemplified by *Model 5.1* in Figure 8. This model retains all parameters of *Model 3*, where incomplete neutralization is consistently observed for ORAL-N clouds, except that, here, all ORAL-N exemplars between iterations 400-490 (~ 23 exemplars) are artificially initialized with a nasality value equal to the mean of the NASAL cloud, plus an additional .15 increase in nasality. As is evident from the trajectory of ORAL-N forms in Figure 8, the nasality of ORAL-N vowels is initialized at .2 and rises normally during the first 400 iterations under the influence of the pre-N production bias. The introduction of highly nasalized exemplars into the ORAL-N clouds is observed by the unusual jump in nasality of ORAL-N exemplars beginning around the 400th iteration. This injection of high-nasality exemplars temporarily shifts individual ORAL-N exemplars toward the nasal end of the continuum but does not durably alter the final mean nasality of the two ORAL-N lexical clouds. Once the intervention ends, the category shifts rapidly back

toward the same incomplete-neutralization endpoint observed in *Model 3*. This is because the encoding strength of the high-nasality exemplars is no different than the other exemplars in the system, so future productions are still drawn toward the optimal nasality value of the objective function.

This result demonstrates that although the phonetic substance of individual exemplars can influence the speed and shape at which a category approaches its stable state, the *value* of that stable state is determined by the relative bias weights encoded in the objective function. In other words, the interaction of production biases dictates the endpoints of neutralization without regard to individual exemplar memories.

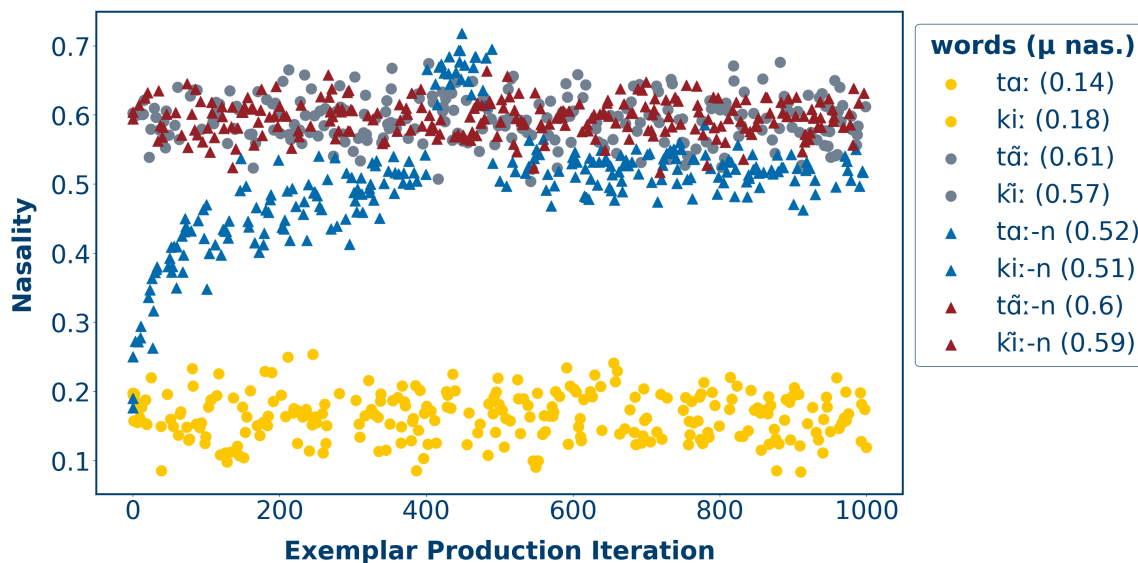


Figure 8: Model 5.1: Injecting high-nasality exemplars into the model does not alter the final stable state.

Indexing the high-nasality exemplars from *Model 5.1* to a ‘linguistic icon’ and encoding these socially elevated exemplar traces more strongly provides one way to counteract the production biases that pull the ORAL-N category toward a final stable state of incomplete neutralization. To implement this socially weighted encoding effect, each exemplar in *Model 5.2* is encoded with a socially determined weight in addition to its mean nasality value. For a typical exemplar, this encoding weight is randomly selected from a normal distribution with $\mu = 1$, $\sigma = .025$. As a result, most exemplars exert approximately equal influence on the production of subsequent exemplars, as in the preceding models.

For the high-nasality ORAL-N exemplars mimicking the linguistic icon, however, an encoding weight of 100 is assigned. The nasality values of these exemplars, therefore, exert an influence on subsequent productions one hundred times greater than the socially ‘typical’ exemplars. Crucially, this means that the high-nasality exemplars introduced between iterations 400–490 no longer function merely as temporary per-

turbations to the exemplar distribution. Instead, their influence persists beyond their relatively small numerical contribution to the exemplar cloud, constititently drawing the nasality of new ORAL-N exemplars toward these socially prominent forms. In this way, socially weighted encoding provides a mechanism through which a small number of socially prominent exemplars can overcome the stabilizing pressure imposed by the competing production biases.

The results of *Model 5.2* are shown in Figure 9. As in *Model 5.1*, ORAL-N exemplars initially increase in nasality under pressure from the pre-N production bias before the high-nasality exemplars are introduced beginning around iteration 400. Unlike *Model 5.1*, however, the increased encoding strength (signified by the enlarged blue triangles) assigned to these exemplars allows their influence to extend beyond the period in which the linguistic-icon forms are directly introduced into the system. Consequently, rather than succumbing to the incomplete-neutralization endpoint established by the objective function, the ORAL-N category continues to produce new exemplars with elevated nasality values, ultimately producing phonetically complete neutralization. Thus, while simply introducing innovative exemplars into the system is insufficient to durably change a category’s stable state, differentially weighting those exemplars based on social information provides an eliminativist mechanism capable of overcoming the pressures of the bias.

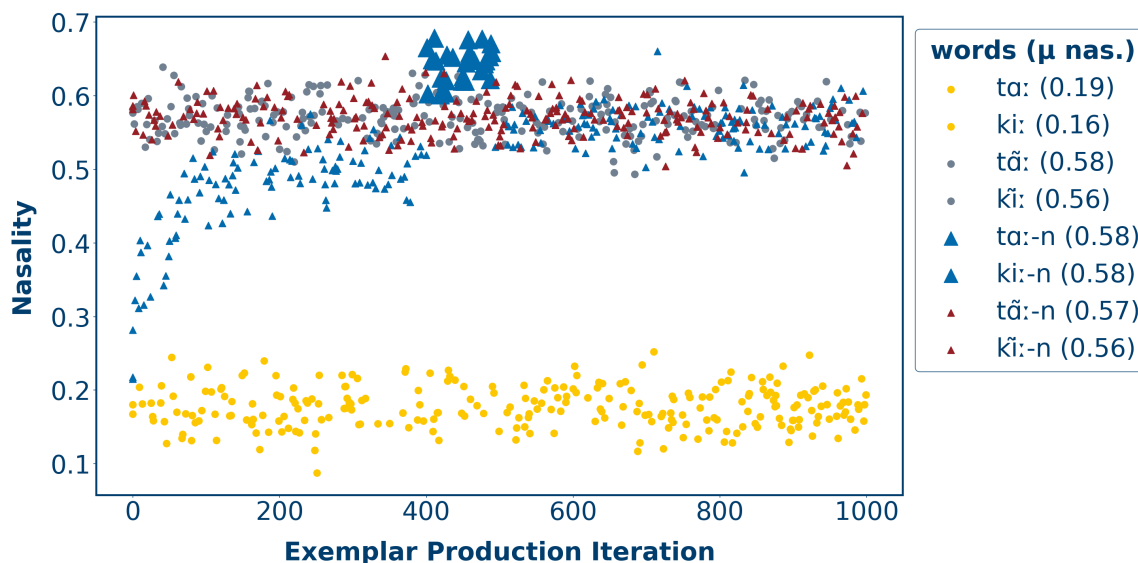


Figure 9: Model 5.2: Adding a socially weighted encoding dimension to high-nasality exemplar values overcomes the pull of the biases to achieve complete neutralization.

The dot and whisker plot in Figure 10 demonstrates the robustness of this result across 100 simulations; by encoding the high-nasality exemplars with a significantly higher weight due to their indexation to a social icon, *Model 5.2* consistently results in the ORAL-N category stabilizing at similar nasality levels to the two NASAL categories.

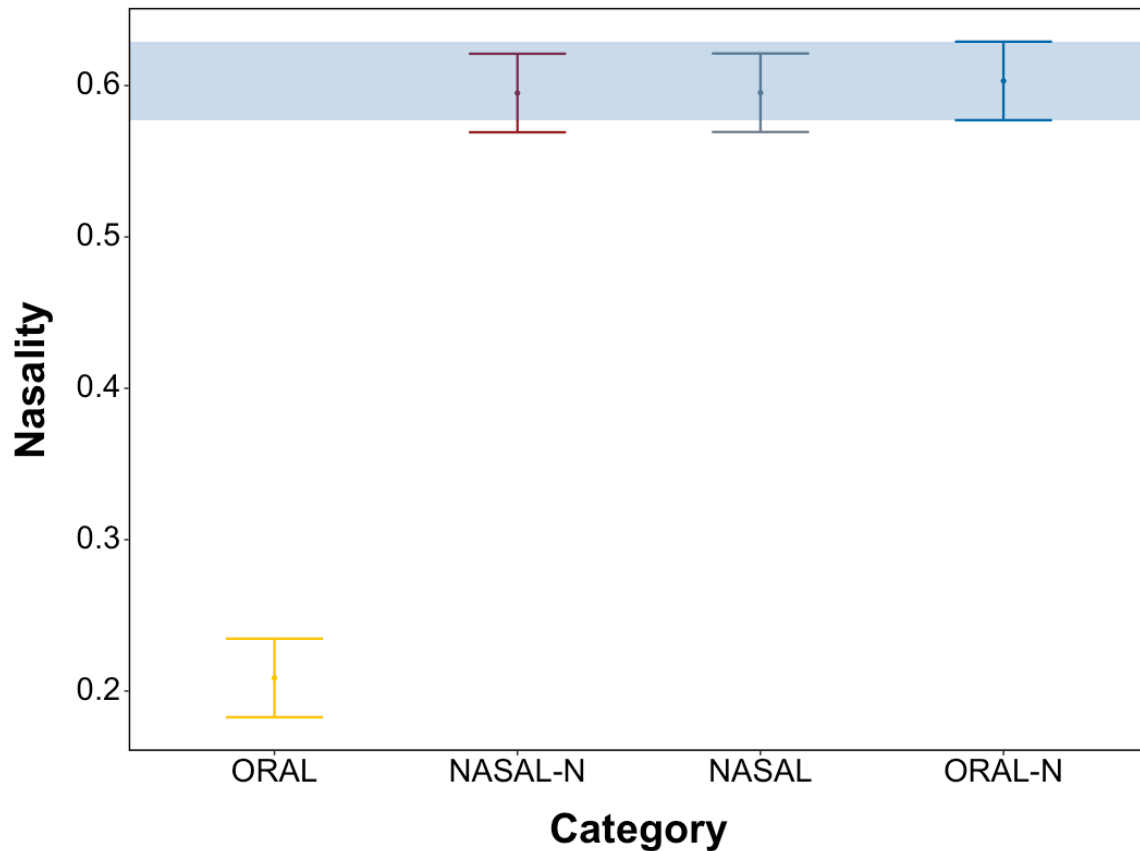


Figure 10: Model 5.2: Adding an appropriately structured socially weighted encoding dimension results in phonetically complete neutralization.

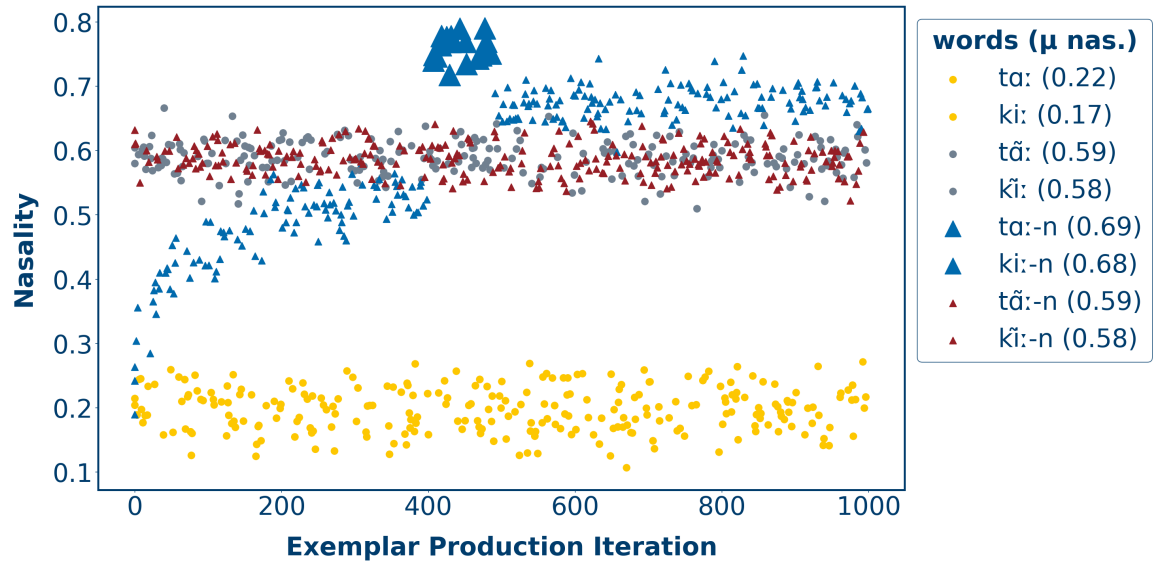
While socially weighted encoding offers an eliminativist method for achieving phonetically complete neutralization, the approach gives rise to at least two issues. The first concerns the necessary persistence of the socially weighted exemplars responsible for shifting the stable state.

Model 5.2 achieves complete neutralization without altering the production-bias mechanism that otherwise draws the ORAL-N category toward an incomplete neutralization endpoint. However, to do so, the high-nasality exemplar memories that exert a disproportionately strong influence on the starting values of subsequent ORAL-N productions must remain resistant to decay. If their relative contribution were instead allowed to diminish through weakening or eventual loss, the unchanged production biases would progressively regain control over new ORAL-N productions, drawing them back toward the incomplete-neutralization endpoint observed in *Model 3* as the high-nasality exemplars decay. Maintaining complete neutralization under this mechanism, therefore, requires a persistently influential set of high-nasality exemplars, whether through prolonged retention or continued replenishment of some kind. Although so-

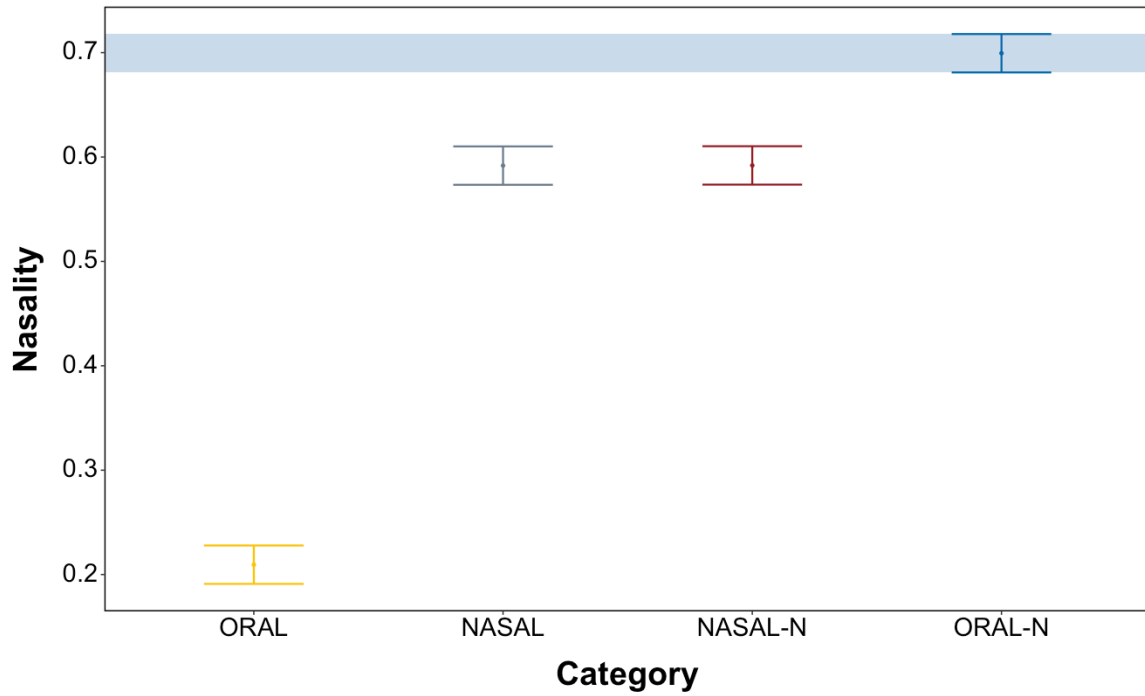
cially salient tokens may be more strongly encoded and better retained than less salient tokens (Sumner et al., 2014), this does not establish that their influence remains sufficient to sustain complete neutralization indefinitely.

A second, and perhaps more pressing, issue concerns the range of neutralization patterns predicted by using socially weighted encoding to manipulate the stabilization point of neutralization. Consider a socially prominent speaker whose ORAL-N productions exhibit unusually high nasality, significantly exceeding that of the NASAL categories of the language user in question. Mimicked exemplars associated with the ‘high-nasality’ speaker receive disproportionately high encoding weights, thereby shifting the starting values of subsequent ORAL-N productions toward these extreme values. If these exemplars are sufficiently extreme, they can draw subsequent productions *above* the NASAL categories. Indeed, nothing in the present implementation requires the resulting endpoint to remain within the bounds of the nasality of the NASAL categories. In other words, the same mechanism under the socially weighted encoding approach that achieves complete neutralization also gives way to *over-neutralization*, in which the neutralizing category stabilizes at a phonetic value more extreme than the category toward which it is converging.

Model 5.3 demonstrates this prediction by increasing the initial nasality of the socially weighted ORAL-N exemplars to .25 above the NASAL mean, while retaining encoding weights centered on 100 and the same production-bias mechanism. As shown in Figure 11a, the socially weighted exemplars pull the ORAL-N category beyond the two NASAL categories, resulting in a mean nasality of .69, compared with .59 and .58 for NASAL-N and NASAL, respectively. Figure 11b confirms that this result is consistent across simulations. Over-neutralization follows from applying the same mechanism responsible for complete neutralization in *Model 5.2* to exemplars with more extreme phonetic values. Crucially, the result conflicts with the typological generalization reported by Du and Durvasula (2025, p.21), who note that no cases have been discovered in which a neutralizing category stabilizes beyond the category toward which it neutralizes. Neutralization is, therefore, not bounded by the NASAL categories under the socially weighted encoding mechanism implemented here. An account based on this mechanism thus requires an additional explanation for what restricts socially driven change to the attested range of neutralization patterns.



(a) Trajectory of an individual simulation.



(b) Summary across 100 simulations.

Figure 11: Model 5.3: Adding a socially weighted encoding dimension overgenerates potential neutralization patterns.

5.3 Model 6 - change in production bias strength

Given the limitations of frequency manipulations and socially weighted encoding, I propose an alternative explanation in which the strength of the relevant production biases varies across languages and/or speakers. Such variation changes the endpoint toward which productions are drawn by the objective function, allowing different degrees of neutralization to arise without requiring the persistent influence of exceptionally weighted exemplars. At the level of an individual speaker, differences in production biases may reflect physical factors, such as vocal tract physiology. Systematic differences across speech communities, however, raise the possibility that speakers acquire language-specific settings governing the relative influence of these biases. In the present framework, this would amount to language-specific knowledge about the relative strength of the global production biases themselves, rather than knowledge about how individual exemplars are encoded or weighted.

Model 6 implements this possibility by increasing the scalar on the pre-N nasality bias from 5 to 25. Strengthening this bias increases its contribution to the objective-function penalty relative to the morphological similarity bias, shifting the optimal production value further toward the mean of the NASAL category. As shown in Figure 12, the ORAL-N forms converge to nasality values similar to those of the NASAL and NASAL-N categories when the pre-N bias weight is increased in this way.

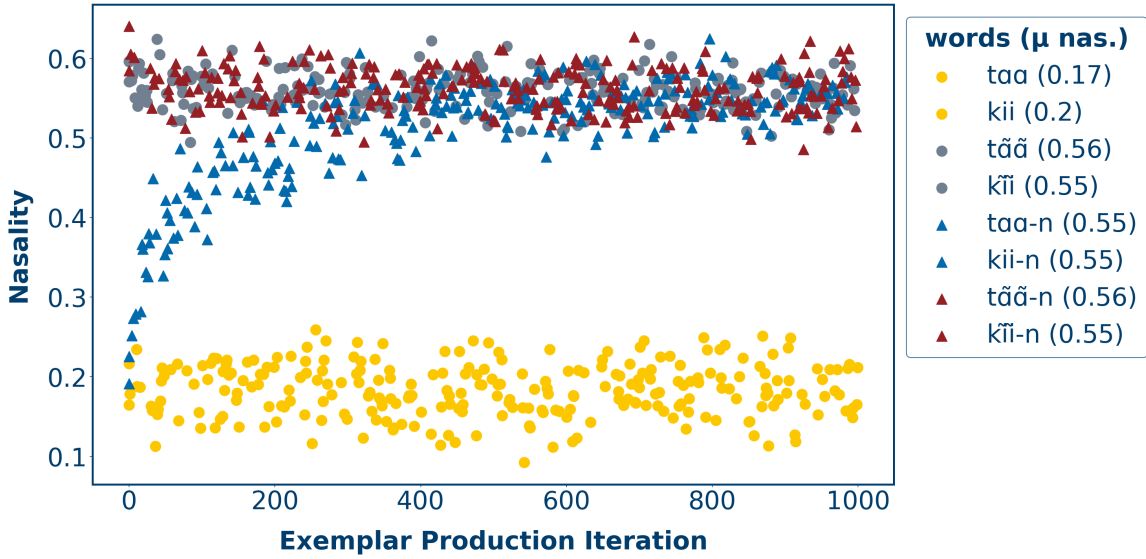


Figure 12: Model 6: selection, entrenchment, pre-N/similarity biases, noise, storage, decay.

The summary across 100 simulations in Figure 13 shows that this near-convergence is reproduced across runs. Crucially, achieving complete neutralization through the modification of bias strengths also constrains the range of possible neutralization endpoints. With the squared penalties implemented here, the optimal production value is

a weighted average of the ORAL base and the NASAL target. This optimum, therefore, necessarily lies between those values. Increasing the pre-N bias strength moves the optimum toward the NASAL target without the ability to move it beyond that target. Random noise may still produce individual tokens with more extreme nasality, but strengthening the bias does not itself establish an over-neutralizing endpoint. This property addresses the overgeneration problem identified in *Model 5.3*: the relative bias strengths determine how closely ORAL-N approaches the NASAL categories while preserving a principled bound on the location of the production optimum.

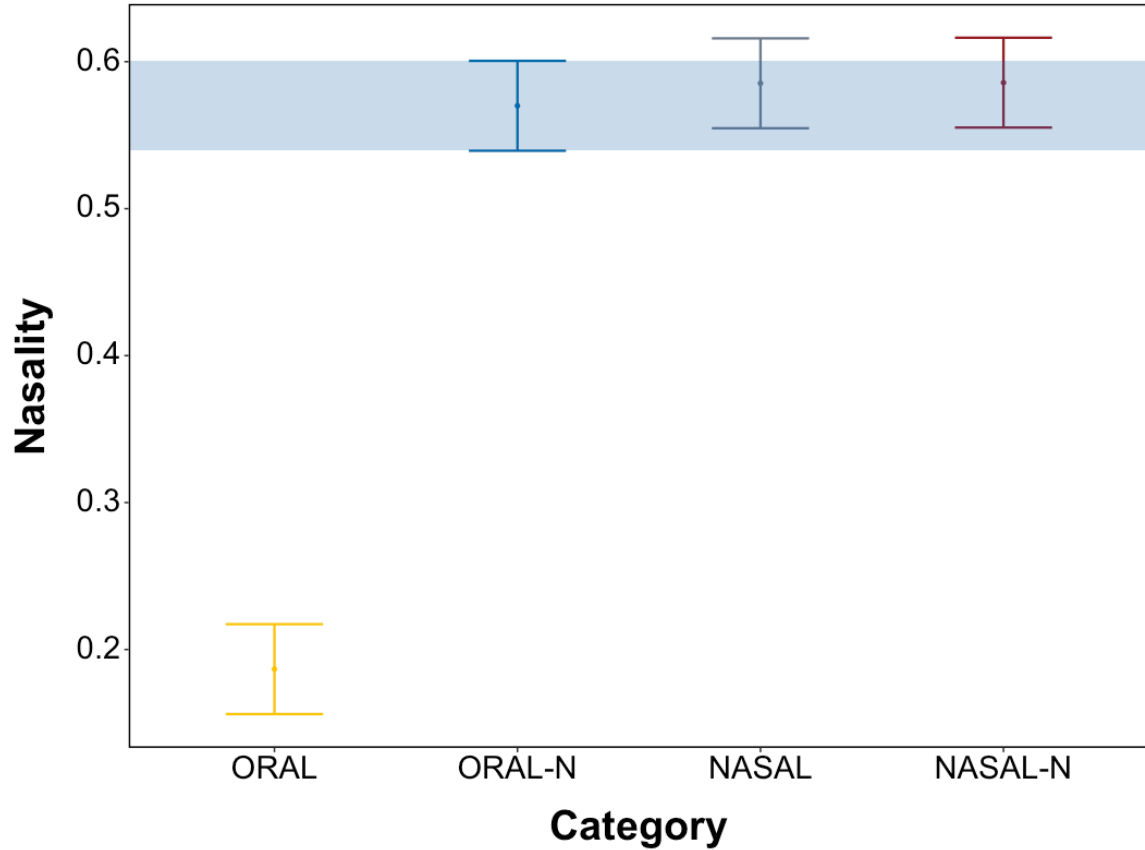


Figure 13: Model 6: selection, entrenchment, pre-N/similarity biases (increased pre-N bias weight), noise, storage, decay.

6 Discussion

The simulations presented here demonstrate that mechanisms available within eliminativist exemplar models *can* generate both incomplete and complete neutralization, but their explanatory adequacy depends on the additional patterns they predict and theoretical machinery they assume. Frequency manipulations assume particular re-

relationships between lexical frequency and the degree of neutralization whose cross-linguistic validity requires empirical support. Socially weighted encoding permits complete neutralization under unchanged production biases, but sustaining this outcome requires the continued influence of exceptionally weighted exemplars, whether through prolonged retention or replenishment. Moreover, the same mechanism predicts over-neutralization when those exemplars have sufficiently extreme phonetic values.

Changing production-bias strength provides another means of regulating the neutralization endpoint while preserving a principled bound on the production optimum. Taken together, these results show that accounting for neutralization requires specifying both the contents of speech memories and the mechanisms governing their influence on production.

Permitting speakers to exert control over production bias strengths, as in *Model 6*, suggests that the knowledge responsible for differences in neutralization is represented at levels beyond individual episodic memories and their relationships to one another. If bias weights differ systematically across languages, they constitute language-specific knowledge beyond the phonetic contents, encoding strengths, and relationships of individual speech representations. Introducing learned production-bias weights thus expands the locus of linguistic knowledge beyond the exemplar network. This challenges the eliminativist commitment that language-specific knowledge resides exclusively in speech representations and their interactions.

The simulations presented in this paper motivate this additional locus of knowledge, but they do not establish how speakers acquire the relevant settings. One possible source of bias weight adjustment is the social context in which speech is exchanged. Social meaning is known to influence the evaluation and uptake of phonetic variants (Eckert and Labov, 2017), and, as mentioned previously, socially salient speech likely receives disproportionately strong encoding in memory (Sumner et al., 2014). *Model 5.2* implements this possibility by assigning exceptionally high encoding weights to high-nasality exemplars associated with a linguistic icon. An alternative implementation mechanism is that exposure to such productions changes the relative importance of the production biases themselves. For example, a speaker encountering highly nasalized ORAL-N forms from a socially influential talker might increase the weight of the pre-N nasality bias in order to bring subsequent productions closer to those of the influential talker. Under this proposal, social experience has a consequence that extends across the forms governed by the bias. Once acquired, the adjusted production setting could continue to shape new productions even after the initiating exemplars decay. This would constitute socially conditioned adjustment of production biases, extending the influence of social information beyond the encoding of individual memories.

Perceptual reinterpretation offers another possible source of bias recalibration. In the hypocorrection account of Ohala (1993), a listener may fail to compensate for the nasalizing influence of a following consonant and instead interpret the resulting

nasalization as a property of the vowel. Within the present framework, one possible production consequence of this reinterpretation would be an increase in the strength of the pre-N nasality bias to match the misperceived increased nasality of the vowel.

Finally, Beddor (2009) provides a further perceptual motivation for bias weight adjustments. Anticipatory vowel nasalization and the following nasal consonant provide related information, allowing listeners to draw on the coarticulatory effect as well as the nasal consonant itself to aid in perception. Changes in the distribution of this information, including a reduction in the duration of the nasal consonant, may be accompanied by greater reliance on vowel nasalization as a perceptual cue. In the present model, a production consequence of this redistribution of cue weighting could be represented by increased weighting of the pre-N nasality bias. Perceptual cue weights and production-bias weights are distinct, however, and the proposed relationship between them is not fully worked out.

All of these explanations for a source of the bias weight change distinguish the experiences that initiate change from the linguistic knowledge that sustains the change's consequences. Social evaluation, coarticulatory misperception, and changes in cue weighting could each contribute to the acquisition of different production settings, either independently or in combination. Regardless of what sources initiate the adjustment in bias weights, a stable adjustment in the weights constitutes a form of knowledge that extends beyond individual speech representations.

References

- Beddor, P. S. (2009). A coarticulatory path to sound change. *Language*, 85(4):785–821.
- Bennett, R. T. and Eischens, B. (2026). "La neutralización incompleta de la nasalidad en el Mixteco: El rol de la abstracción" [Incomplete neutralization of nasality in Mixtec: the role of abstraction]. Presentation at Sound Systems of Latin America (SSLA) 5.
- Braunschweiler, N. (1997). Integrated cues of voicing and vowel length in German: A production study. *Language and Speech*, 40(4):353–376.
- Braver, A. (2013). *Degrees of incompleteness in neutralization: paradigm uniformity in a phonetics with weighted constraints*. PhD thesis, Rutgers University.
- Braver, A. (2014). Imperceptible incomplete neutralization: production, non-identifiability, and non-discriminability in American English flapping. *Lingua*, 152:24–44.
- Braver, A. (2019). Modelling incomplete neutralisation with weighted phonetic constraints. *Phonology*, 36:1–36.

- Braver, A. and Kawahara, S. (2016). Incomplete neutralization in Japanese monomoraic lengthening. In Albright, A. C. and Fullwood, M. A., editors, *Proceedings of the Annual Meeting on Phonology*, volume 2.
- Burzio, L. (2000). Segmental contrast meets output-to-output faithfulness. *The Linguistic Review*, 17:367–384.
- Bybee, J. (2001). *Phonology and language use*. Cambridge: Cambridge University Press.
- Chomsky, N. and Halle, M. (1968). *The Sound Pattern of English*. Harper and Row, New York.
- Coretta, S. (2020). *Vowel duration and consonant voicing: A production study*. PhD thesis, University of Manchester.
- Dinnsen, D. A. and Charles-Luce, J. (1984). Phonological neutralization, phonetic implementation and individual differences. *Journal of Phonetics*, 12:49–60.
- Du, N. and Durvasula, K. (2022). Phonetically incomplete neutralisation can be phonologically complete: evidence from Huai’an Mandarin. *Phonology*, 39(4):559–595.
- Du, N. and Durvasula, K. (2025). *Psycholinguistics and Phonology: The forgotten foundations of Generative phonology*. Cambridge Elements in Phonology. Cambridge: Cambridge University Press.
- Eckert, P. (2000). *Linguistic variation as social practice*. Blackwell, Oxford, UK.
- Eckert, P. and Labov, W. (2017). Phonetics, phonology, and social meaning. *Journal of Sociolinguistics*, 21(4):467–496.
- Ellis, L. and Hardcastle, W. J. (2002). Categorical and gradient properties of assimilation in alveolar to velar sequences: evidence from epg and ema data. *Journal of Phonetics*, 30(3):373–396.
- Englert, E. G. (2022). A first look at Mankiyali morphology. Master’s thesis, University of North Texas.
- Gerfen, C. (2002). Andalusian codas. *Probus*, 14:247–277.
- Kaplan, A. (2015). Positional neutralization in an exemplar model: The role of unique inflectional bases. In *Proceedings of the 2014 Annual Meeting on Phonology*.
- Kim, H. and Jongman, A. (1996). Acoustic and perceptual evidence for complete neutralization of manner of articulation in Korean. *Journal of Phonetics*, 24:295–312.

- Kochetov, A. and Pouplier, M. (2008). Phonetic variability and grammatical knowledge: an articulatory study of korean place assimilation. *Phonology*, 25(3):399–431.
- Lahiri, A., Schriefers, H., and Kuijpers, C. (1987). Contextual neutralization of vowel length: Evidence from Dutch. *Phonetica*, 44:91–102.
- Matsui, M. (2015). *Roshia-go ni okeru yuusei-sei no tairitsu to tairitsu no jyakka: onkyo to chikaku*. PhD thesis, Hiroshima University. Voicing contrast and contrast reduction in Russian: acoustics and perception.
- Morley, R. (2023). *Sound structure and sound change: A modeling approach*. Number 4. Language Science Press, Berlin.
- Myers, S. and Padgett, J. (2014). Domain generalisation in artificial language learning. *Phonology*, 31(3):399–433.
- Ohala, J. J. (1993). Coarticulation and phonology. *Language and Speech*, 36(2-3):155–170.
- Paramore, J. C. (2026). *From abstract to episodic: Representations in speech production*. PhD thesis, University of California, Santa Cruz.
- Pierrehumbert, J. B. (2001). *Exemplar dynamics: Word frequency, lenition and contrast*, pages 137–157. Amsterdam: John Benjamins.
- Port, R. F. and O’Dell, M. L. (1985). Neutralization of syllable-final voicing in german. *Journal of Phonetics*, 13:455–471.
- Sumner, M., Kim, S. K., King, E., and McGowan, K. B. (2014). The socially weighted encoding of spoken words: a dual-route approach to speech perception. *Frontiers in Psychology*, 4:1015.
- Vietti, A., Alber, B., and Vogt, B. (2018). Initial laryngeal neutralisation in Tyrolean. *Phonology*, 35(1):79–114.
- Warner, N., Jongman, A., Sereno, J., and Kemps, R. (2004). Incomplete neutralization and other sub-phonemic durational differences in production and perception: evidence from Dutch. *Journal of Phonetics*, 32:251–276.
- Wedel, A. B. (2004). *Self-organization and categorical behavior in phonology*. PhD thesis, University of California, Santa Cruz.
- Yu, A. C. L. (2007). Understanding near mergers: the case of morphological tone in Cantonese. *Phonology*, 24:187–214.
- Zeng, Y., Wang, C., and Zhang, J. (2025). Cascading activation in spoken word production drives incomplete neutralization: An internet-based study of Mandarin 3rd tone sandhi. *Journal of Phonetics*, 112.